

Module 3 – Databases & APIs

1. MongoDB Introduction

What is MongoDB?

MongoDB is a NoSQL database used to store data in flexible, JSON-like documents.

Why MongoDB?

- Schema-less structure
- High performance
- Easy scalability
- Stores data in documents

Key Concept

Data is stored in collections instead of tables.

2. Mongoose ODM

What is Mongoose?

Mongoose is an Object Data Modeling (ODM) library for MongoDB and Node.js.

Why Mongoose?

- Defines schemas
- Validates data
- Simplifies database queries

Example:

```
const mongoose = require("mongoose");

const userSchema = new mongoose.Schema({
  name: String,
  age: Number,
});

const User = mongoose.model("User", userSchema);
```

3. CRUD Operations

What is CRUD?

CRUD stands for Create, Read, Update, Delete operations.

Operations Explained

- Create → Add new data
- Read → Get data
- Update → Modify data
- Delete → Remove data

Example:

```
User.create({ name: "Ali", age: 22 });  
  
User.find();  
  
User.updateOne({ name: "Ali" }, { age: 25 });  
  
User.deleteOne({ name: "Ali" });
```

4. REST APIs

What is REST API?

REST API is a way for applications to communicate over HTTP using standard methods.

HTTP Methods

- GET → Fetch data
- POST → Create data
- PUT → Update data
- DELETE → Remove data

Example:

```
app.get("/users", (req, res) => {  
  res.send("Get Users");  
});
```

5. API Testing

What is API Testing?

API testing checks if APIs are working correctly.

Tools Used

- Postman
- Insomnia
- Thunder Client

What is Tested

- Response status
 - Data accuracy
 - Performance
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6. Key Concepts to Remember

- MongoDB stores data in documents
 - Mongoose helps manage MongoDB easily
 - CRUD operations manage data lifecycle
 - REST APIs connect frontend and backend
 - API testing ensures correctness
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7. Summary

Databases and APIs form the core of backend development. MongoDB stores data, Mongoose simplifies interaction, CRUD operations manage data, and REST APIs enable communication between systems.